ANKIT KUMAR | 25AG62R05

LAND AND WATER RESOURCE ENGINEERING

+91-6205201375 | [ankit_200325@kgpian.iitkgp.ac.in] | [LinkedIn] | [Portfolio] | [GitHub]

EDUCATION

Year	Degree/Exam	Institute	CGPA/Marks
2027	M.Tech, Land & Water Resource Engineering	IIT Kharagpur	7.14/10
2025	B.Tech, Agricultural Engineering	Anand Agricultural University	8.05/10
2021	Higher Secondary (XII)	Bihar School Education Board	83.8%
2019	Secondary (X)	Central Board of Secondary Education	89.4%

INTERNSHIPS / EXPERIENCE

Research Intern | Indian Institute of Tropical Meteorology (IITM), Ministry of Earth Sciences — Pune **[May'26-Jul'26]**

Deep Learning | Satellite Data Pipelines | Statistical Modelling | HPC — 3 research projects

- **Deep Learning Reconstruction of Radar Composite Reflectivity (CREF) from INSAT-3DR Geostationary Imagery:** comparative study of U-Net, GAN-refined, and Transformer architectures against C-Pol radar over the monsoon core zone (ART-Bhopal).
- **Physics-informed CREF Retrieval:** deep learning retrieval with calibrated diffusion downscaling; model evaluation, uncertainty analysis, and explainability with pre-registered experiment design.
- **NIRA Dashboard:** near-real-time INSAT–radar assimilation and CREF visualisation platform; built end-to-end Python pipelines for satellite and radar data on the Arka high-performance computing facility.

Research Intern | EXPERIQS PRIVATE LIMITED (IIT Bombay-based startup) — Hybrid **[Jan'26-Jun'26]**

GeoAI | Soil Property Prediction | Climate Risk Assessment | Hydrological Modelling

- Developed GeoAI and machine learning solutions for climate risk assessment and geospatial analytics, including AI-driven platforms for **soil property prediction** and flood risk intelligence using Sentinel satellite imagery and Google Earth Engine.
- Delivered a site-level **Physical Climate Risk Assessment (CII PCRAF, TCFD-aligned)** for a Godrej & Boyce plant: hazard-exposure-vulnerability modelling of flood, heat, drought and landslide risk with expected-annual-loss and value-at-risk quantification.
- Executed HEC-HMS rainfall-runoff and 2D HEC-RAS unsteady-flow simulations (2.5 m DEMs, IMERG rainfall), integrated Sentinel-1 SAR and Sentinel-2 indices for flood/landslide risk zoning, and produced hazard maps in ArcGIS Pro.

Research Intern | Sardar Patel Renewable Energy Research Institute (SPRERI) — Gujarat **[Aug'24-Nov'24]**

- Performed time-series thermal analysis of nanofluids (optimal 10% concentration) and DSC thermodynamic analysis of $NaNO_3$/KNO_2 mixtures (60:40 ratio, 92.79 J/g latent heat) for solar thermal energy storage.

PROJECTS

[HydroRisk Atlas – Flood Risk Intelligence Platform] **[Feb'26-Present]**

Python | Google Earth Engine | Machine Learning | Streamlit | GIS

- Built a satellite-based flood risk platform using Google Earth Engine and Python: Sentinel-1 SAR change detection and Sentinel-2 indices (NDVI, NDWI) for susceptibility mapping and inundation detection.
- Trained and validated Random Forest, XGBoost, and LightGBM models for flood risk prediction; deployed an interactive Streamlit dashboard for geospatial visualisation and risk assessment.

[Bihar Hydro-Climatic Risk Atlas – Flood & Groundwater Risk DSS] **[Dec'25]**

Python | Google Earth Engine | ML | Time-Series | Streamlit

- Built a satellite-driven hydro-climatic risk platform with Random Forest models assessing flood pressure and groundwater stress; designed a Compound Risk Index for co-occurring hazards.
- Implemented SHAP explainability and time-series analysis with an interactive Streamlit GIS dashboard for decision support.

[Open-Source Tools — GitHub]

- **WRF_Python_Scripts**: WRF model visualisation suite (synoptic/mesoscale, precipitation, fire weather, tropical products); **precip-index**: SPI/SPEI computation for climate-extremes monitoring; **Ai_forestDetection**: ML-based forest cover detection.

PUBLICATIONS & MANUSCRIPTS

- **First author:** "Deep Learning Reconstruction of Radar Composite Reflectivity from INSAT-3DR Geostationary Imagery: A Comparative Study of U-Net, GAN-Refined, and Transformer Architectures" — **International Journal of Climatology (Special Issue)**, communicated Aug 2026 (IITM Pune).
- "Physics-informed CREF Retrieval from INSAT-3DR over Central India with Calibrated Diffusion Downscaling to 1 km" — manuscript in preparation (IITM Pune).
- **Co-author:** "Coupled Hydrologic-Hydraulic Modelling and Basin-wide Susceptibility Screening of Flood Hazard and Road Disruption in the Upper Beas Basin" — manuscript under revision.
- "Ultrasonic-Guided Bluetooth-Controlled Spraying Rover for Precision Agriculture" — [Journal of AgriSearch (NAAS 4.95), published].

SKILLS AND EXPERTISE

Programming & Data Pipelines: Python (Pandas, NumPy, Scikit-learn, Matplotlib, Plotly), Git, Docker, Streamlit, Folium, automated geospatial data pipelines, HPC

Statistical & Machine Learning: Statistical modelling, Random Forest, XGBoost, LightGBM, ANN, CNN, U-Net, GAN, Transformer, SHAP explainability, time-series analysis, probabilistic risk assessment

Geospatial & Remote Sensing: Google Earth Engine, ArcGIS Pro, QGIS, SNAP, Sentinel-1 SAR, Sentinel-2, INSAT-3DR, IMERG, DEM analysis, HEC-RAS (2D), HEC-HMS, WRF

Key Subjects: Digital soil mapping, Climate change and water resources, Statistical and ML methods in water resources, Geo-informatics for land & water resources, Probabilistic risk assessment, AI applications in agriculture, Surface water hydrology

ACHIEVEMENTS, CERTIFICATIONS & CONFERENCES

- **Grand Finalist (Top 34 of 15,104 teams) — Bharatiya Antariksh Hackathon (BAH) 2026, ISRO–NRSC, Hyderabad**: Generative AI-based cloud removal and reconstruction for LISS-IV satellite imagery; both registered teams shortlisted.
- **[National Finalist (Top 38) — MoPR AI/ML Hackathon (IIT Tirupati Navavishkar i-Hub Foundation)]**: AI/ML-based DTM creation from point-cloud data and drainage network development.
- **GATE 2025 AIR-69 & GATE 2024 AIR-111** (Agricultural Engineering); ICAR National Talent Scholarship; IIRS–ISRO certifications: AI/ML for Geodata Analysis (Aug 2024), Remote Sensing & Digital Image Analysis (Oct 2024).
- **Conference talks:** "IoT-Based Automated Fertigation System" (~38.5% water savings) — 35th National Convention of Agricultural Engineers (IEI), JNKVV; thermal energy storage research — 58th ISAE Annual Convention, Parbhani (Nov 2024).